Sensory Integration

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Introduction

Sensory integration, or sensory processing, is the organisation of sensation for use.[1] It refers to the ability of our brain to recognise and respond to signals sent by our sensory system. Our senses include hearing, vision, smell, taste, touch, proprioception, interoception, and our vestibular system - which provides information about movement, changing head position and gravity.

Sensory integration plays an important role in a child's development. This includes their ability to develop and maintain social-emotional relationships, as well as develop motor skills, cognitive skills, adaptive skills, and others.[1] Children can be hypo- or hyper-sensitive to sensory inputs. When children have difficulty processing and responding to sensory information, their ability to participate in daily activities, school activities, etc. can be affected.

Sensory integration (SI) has been described by Amy Stephens (director of Lifelong Learning) as a framework of reference to be used across the multidisciplinary team and as both an intervention and approach. The end point of SI is to allow the individual to identify what they need from their environment, and how they can manipulate their environment for what they need.[2]

This article provides a brief overview of sensory integration and offers recommendations for sensory integration therapy for children with cerebral palsy when sensory processing challenges are present. Optionally, you might like to read Sensory Integration Therapy in Paediatric Rehabilitation to learn more about the development, conditions treated, and the role of the Occupational Therapist.

Senses

While our five senses of touch, vision, hearing, smell and taste are commonly known, there are other senses which inform us about our bodies and the world they inhabit.[3]

Our senses can be grouped into three areas:[3]

1. **Exteroception** - sensation from outside the body: Making use of the five common senses
2. **Proprioception** - sensation about body position and movement
3. **Interoception** - sensation from inside the body, which is perceived through organs and viscera

The following sections provide a brief description of the systems most commonly addressed and utilised in a therapeutic environment.

Somatosensory system

The somatosensory system comprises of touch and proprioception.[4]

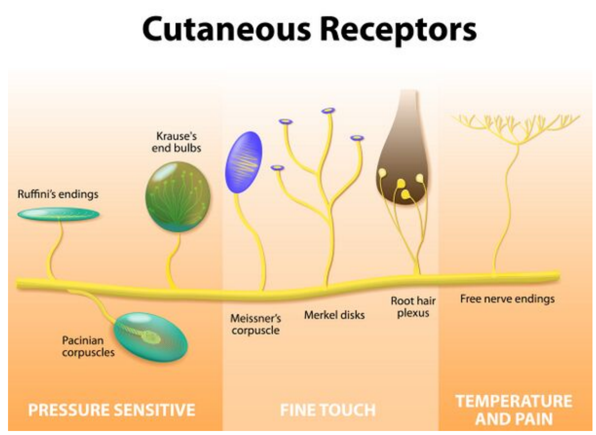
One of the most important functions of this system is praxis: being able to plan and execute tasks.[4]

1. Touch / Tactile System

The tactile system is the sense of touch. It is the sense which allows for the perception, organisation and integration of information through receptors on our skin.[5]

Information is received from receptor cells in the skin.

> **Cutaneous** receptors provide information about light touch, pressure, vibration, temperature, and pain
>
> **Hair follicles**: also effect touch perception
>
> **C-fibre low threshold mechanoreceptors (LTM)** respond to "pleasant" and "effective" mechanical stimuli like gentle stroking and brushing and small changes in skin temperature[11]
>
> ***Thermoreceptors*** detect changes in temperature



Cutaneous Receptors

The receptors in the skin relay information to different areas of the brain, namely the primary sensory cortex and the secondary somatosensory cortex.

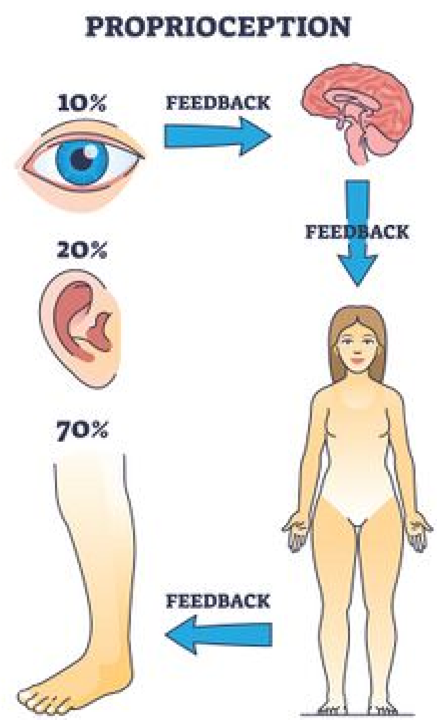
Further somatosensory-vestibular-visual integration of information occurs in regions such as the vestibuli nuclei, thalamus and cortex.[4]

The tactile system has multiple functions which range from the simple reflex withdrawal, to complex integration with other senses.[4] Information received by the primary motor cortex is important for object manipulation and grasp, while information going to the secondary somatosensory region is involved in linking present and past sensations, needed in motor planning (a component of praxis). Tactile information is important in this process in regard to feedforward processing and prediction of movement.[4]

2. Proprioception

Proprioception (kinaesthesia) helps us to move. Proprioception refers to information arising from skin, muscles, joints, ligaments, and bones.[1] It allows us to perceive the location, movement, and actions of our body.[12] The sense of effort, the sense of force and the sense of heaviness are also all involved in proprioception:[13]

- when we change position or move our limbs, the tissues (e.g. skin, muscle, tendons, etc.) surrounding the moving joint are deformed[12]
- it has been suggested that muscle spindles "play the major role in kinaesthesia"[12]
- the skin receptors provide additional information[12]
- it is also suggested that Golgi tendon organs contribute to proprioception[12]



Proprioception

3. Vestibular System

The vestibular system provides information about movement, gravity, and changing head position.[1] It contains two receptor types:[4]

- **semicircular canals** - detect angular movement of the head
- **otolith organs** - detect linear movement of the head and the pull of gravity

The vestibular system is linked to many regions and structures of the brain, including the cerebellum, cranial nerves and even the cortex. This means that it plays a role in many functions of the brain and behaviours.[4]

One important function is arousal regulation. Rapid or unpredictable acceleration is associated with increased alertness, while slow, rhythmical movement is associated with calming and drowsiness.[4]

Our neck and trunk musculature, important for head and postural control, is affected by vestibular input via the vestibulospinal tract. The refining of postural control is also affected by vestibular information coming to the cerebellum via the brainstem. Lane et al.[4] highlighted the contribution the vestibular system has as a bilateral system, linking deficits in vestibular information to poor equilibrium reactions.

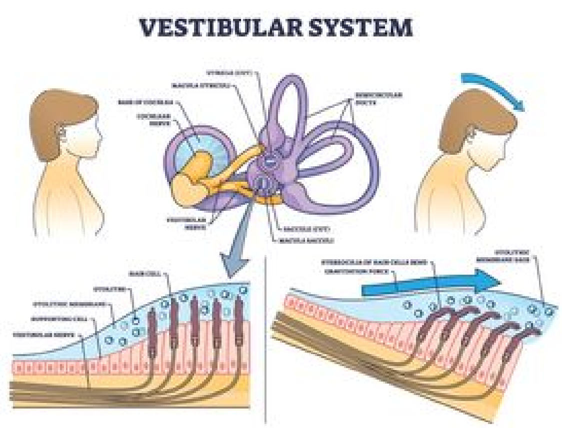
Cranial nerves receiving vestibular information via the longitudinal fasciculus aids eye and head movements. This information provides stability to an individual's surroundings, even when the person is moving.[4]

Through connections to the visual and proprioceptive systems, vestibular information can contribute towards anticipatory motor actions and plans.[4]

In summary the vestibular system:

- affects arousal regulation
- affects postural control
- facilitates balance and equilibrium responses
- affects bilateral co-ordination
- aids in maintaining a stable visual field (helps to stabilise our eyes when we are moving)
- aids in spatial perception (telling us where we are in space and if the objects around us are moving or stationary)

To learn more about the peripheral vestibular system, please click here.



Vestibular System

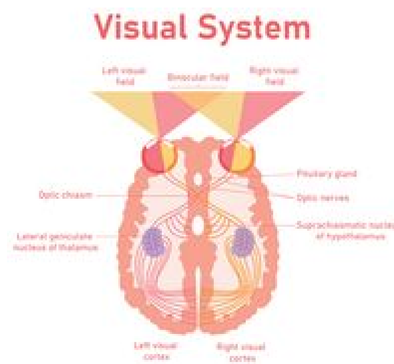
4. Hearing / Auditory System

- The auditory system processes sound in the environment[1]

- Auditory receptors in the inner ear identify various sound stimuli (e.g. loud / soft, far / near)[1]
- These sound stimuli are then processed by the central nervous system to determine an appropriate response:
    - for example, in situations where safety is a concern (e.g. a fire alarm goes off), we recognise the sound and act accordingly)[1]
- Posture control can be influenced by sound frequency[14]

5. Visual System

- Our visual system helps us see and perceive the environment around us[1]
- It identifies sights and understands what the eyes see[1]
- Visual inspection is also important in maintaining body balance as it helps position the body in space[15]

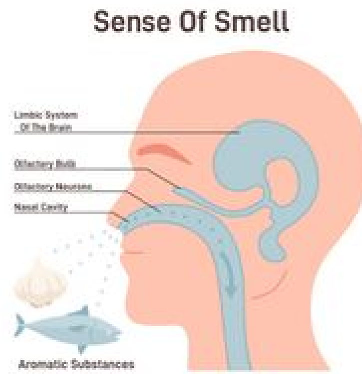


Visual System

6. Smell / Olfactory System

- Our olfactory system allows us to distinguish various odours in the environment[1]
- The three olfactory functions include:
    - protection from environmental hazards: identifying safe or potentially dangerous situations (e.g. the smell of smoke)[1][16]
    - nutrition: contributing to our appetite drive and food intake regulation
    - social behaviour: human sweat, detected by our olfactory system is a powerful chemo-communicator and plays a role in fostering relationships; smell is also involved in emotional regulation[16]

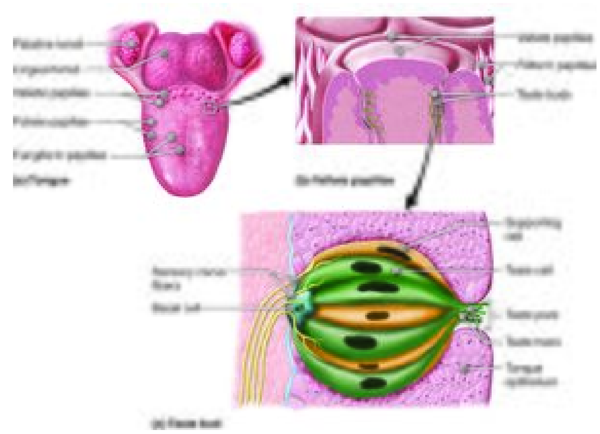
> It is important to note that the olfactory system is closely linked to the limbic system, which is responsible for emotions, behaviours and memories.[16]



Sense of Smell

7. Taste / Gustatory System

- Our gustatory system distinguishes four tastes: sweet, sour, salty, and bitter[1]
- It is closely associated with the olfactory system
- It allows us to identify desirable foods that are pleasant to us, as well as those that are potentially undesirable, such as a bitter dish[1]



Tongue and Taste Buds

8. Interoception

Interoception is "the sense involved in the detection of internal regulation, such as heart rate, respiration, hunger, and digestion".[17] Through interoception, we can distinguish our body's internal sensations.[18]

> Sensory integration or sensory processing "is the potential to develop adequate motor and behavioural reactions to stimulus."[19] -- Ayres

Sensory integration involves input from the senses which is *received*, *organised* and *interpreted* to create a reaction appropriate to the stimuli received.[4]

In 1960, Ayres proposed that:[20]

1. learning takes place as a function of reward and reinforcement
2. one learns what one does
3. learning takes place because there is a purpose for it's taking place

Ayres further proposed that **development of the body scheme, development of posture and motor tasks, and even academic skills** all need the input of **sensory information**. In further regard, she emphasised how the sensory systems develop in relation to and because of one another.[20]

Sensory integration theory places emphasis on "the active, dynamic sensory-motor processes that support movement as well as interaction within social and physical environments."[4]

A great guide to Sensory Integration and Sensory Processing terms can be found in the STAR Institute Sensory Integration & Processing Jargon Guide.

Sensory Integration / Processing Challenges

> Sensory integration dysfunction is a problem in the ability to ‘‘organise sensory information for use.’’[21] -- Ayres

Sensory function is essential for motor ability, social skills, and various behaviours. Adequate sensory integration allows for the foundation of adaptive behaviour. Disruption in **modulation, discrimination** or **integration** of sensory input can create a cascading effect that impacts all levels of sensory processing. This can include motor, social and behavioural components (think 'impairments' and 'activity' levels of the ICF framework). This disruption can translate to problems with participation at home, school, and in the community.[17][4]

In general, sensory integration dysfunction may cause an individual to have difficulty with the following:[17]

- initiating or sustaining peer interactions
- developing relationships
- participating in activities of daily living
- regulating arousal behaviours
- language development

Sensory integration processing deficits may cause an individual to:[17]

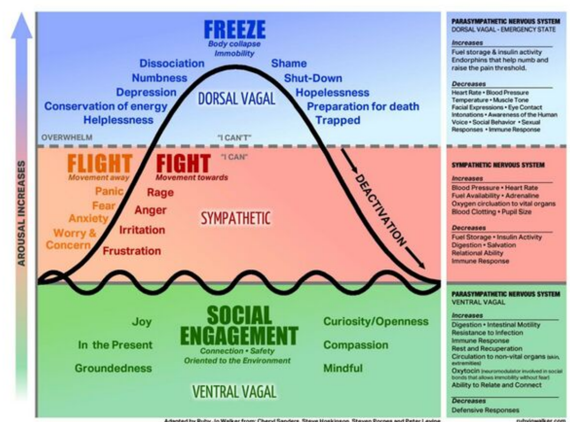
- respond to stimulation more quickly
- respond to stimulation more intensely

- respond to stimulation for longer than individuals who do not have difficulty with sensory integration processing

Examples of behaviour which may be indicative of sensory processing deficits:[17]

- extreme responses to stimuli, including noise in a classroom, odours in a restaurant, the touch of clothing, or the movement of playground equipment.
- "fight, flight or freeze" behavioural responses, such as aggression, withdrawal, or preoccupation with the expectation of sensory input
- severe difficulty forming and maintaining peer relationships
- extreme efforts to control events in the environment by over-reliance on routines
- behaviour regulation problems like temper tantrums, outbursts, hitting, kicking, biting, or spitting
- profound withdrawal from the group.
- slow to respond to sensation, requiring "more intense stimuli to respond to the demands of the situation"[17]

The polyvagal/ nervous system chart is a visual depiction of the levels of arousal which our nervous systems fluctuate between. For learning and engagement in a task, the nervous system needs to be at an appropriate level of arousal i.e. **a regulated state**. If there is too much stimulation via the the sensory system, an individual can be moved into a 'sympathetic' state - a state of stress. Further up on the arousal scale, an individual will enter the 'dorsal vagal' state where there is a complete shut out of stimuli and inability to engage at all.



Polyvagal/nervous system chart

Sensory Deficits and Cerebral Palsy

Children with cerebral palsy often have deficits in one or more sensory systems, including proprioception, tactile sensation, and visual perception.[1] They can also have impairments in *processing* and *modulating* multisensory information, which then affects their activity level, muscle tone, proprioception and emotional responses.[22]

This can affect their functional activities, especially bilateral arm activities, such as eating, playing, dressing, and showering.[23] A child with cerebral palsy and sensory processing difficulties might react to sensory stimuli in the following ways.

- **Hyper-responsiveness to tactile input**:[1]
    - does not like to be touched
    - avoids activities that involve getting messy
    - resists light touch
    - avoids certain types of clothing
- **Hypo-responsiveness to tactile input**:[1]
    - lacks the ability to localise touch or respond when touched
    - places items in their mouth
    - may have a preference for activities or situations which involve brushing their hair, touching or hugging
    - may fail to recognise when their hands or face are messy
    - enjoys activities involving vibration
- **Hyper-responsiveness to proprioceptive input:**[1]
    - cries when they are in weight-bearing positions or when their joints are moved
    - chooses not to move or engage in activities
- **Hypo-responsiveness to proprioceptive input**:[1]
    - bites or chews on non-food objects
    - engages in pinching or hitting others or themself
    - has difficulty changing body posture to match activity demands
    - may have low, high, or variable muscle tone, which impacts the processing of proprioceptive information
- **Hyper-responsiveness to vestibular input:**[1]
    - overreacts when moved in space
    - becomes fearful of bouncing or swinging
    - dislikes sudden or quick movements
- **Hypo-responsiveness to vestibular input:**[1]
    - enjoys being moved and rocked passively
    - seeks opportunities to fall without regard to safety
    - likes excessive spinning, swinging, and active movements

With children who are diagnosed with cerebral palsy, it is important to consider how they express or demonstrate their responsiveness to stimuli. A child may be able use words to tell you when they like or don't like something, but a non-verbal child may communicate differently, especially when they are catagorised at a low GMFCS functioning level. A child may use sounds to communicate interest or level of alertness. Often, a child who is overstimulated, may become louder. However, they may also become quiet and inattentive as a form of 'shutdown'.

Also, look for indicators such as tone change; does the child get stiffer when they engage in the input? Or do their movements change? For example, a child with dyskinesia may have movements which increase in amplitude. A child may also fix certain joints, such as their hips or knees, or demonstrate greater 'posturing'. Other indicators include drooling (an increase in arousal in any system may cause this), or a change in visual attention within the activity.

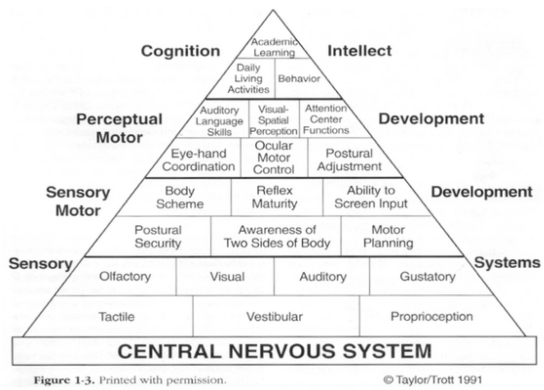
Sensory Integration Therapy

Sensory integration therapy (SIT) is done by trained occupational therapists, physiotherapists and speech therapists, depending on the country and training required.

The "hallmark" of SIT is that it is performed in the context of play. It should include activities that the child enjoys, which ensures that the activities are their own reward.[20]

SIT targets seven sensations: auditory, visual, gustatory (taste), olfactory (smell), somatosensory (proprioception and touch), vestibular, and interoception.[17] It is suggested that SIT directly improves attentional, emotional, motor, communication, and/or social difficulties.[24]

The Pyramid of Learning is an illustration that depicts the foundational skills needed by a child and the skills which then 'build on top' of these. The base needs to be adequately addressed for successful development of the ascending tiers. This is known as the bottom-up approach.[25]



Pyramid of Learning

> The goals of SIT are:[1]
>
> - to facilitate a child's daily functioning
> - to elicit a child's adaptive response in the form of an appropriate reaction to environmental or situational requirements

The Just Right Challenge - activities performed offer a challenge, but are always orientated for success.[26]

The Adaptive Response - the individual or child will adapt their behaviour with new and useful strategies in response to the therapeutic challenge.[26]

Active Engagement - the individual / child will actively participate in the environment and challenge developed by the therapist.[26]

Child Directed - the therapist constantly observes behaviour and behavioural cues in the session. These cues provide leads to creating enticing and sensory-rich activities. The therapist follows the lead and suggestions of the individual / child.[26]

Some recommendations when applying SIT include the following:[1][27]

- establish clear goals - these should be measurable goals focused on the occupations and participation levels
- the most important focus should be the occupations identified by the child and family - *note that occupation as defined in the therapeutic realm is anything which occupies one time in a meaningful way*
- make use of psychometrically sound outcome measures
- clearly explain the state of the evidence to the family, for an informed choice
- ensure the child's physical safety
- prepare the child before starting an activity (safety, posture, muscle tone)
- promote sensory-enriched activities when appropriate (reduce sensory input when needed)
- collaborate with the child on activity choice and maximise the child's success
- guide self-organisation and support optimal arousal
- build a therapeutic alliance through positive and supportive relationships
- provide a "just-right" challenge
- determine the activity's intensity and duration
- promote positive experiences and respect a child's preferences
- consider multiple hypotheses in the occupational analysis for the sensory difficulty
- integrate into daily routine (like chores and art)
- involve the family and suggest changes to tasks and environment
- incorporate every movement into play

Every **environment** a child resides in is appropriate for SIT, including schools, homes, yards, and playgrounds. However, sensory rooms tend to be utilised most frequently for treatment. Ensuring a child's physical safety is essential. Clinicians must carefully observe a child's reactions and pay attention to signs the child is becoming overwhelmed. If the child does become overwhelmed, the clinician should take prompt action and proactively change the situation.[1]

Equipment: a sensory room should include at least three inventories of working materials for each sensation:[1]

Note: When introducing foods or liquids it is important to be aware of any allergies or food sensitivities. In addition, any loss of or impaired sensation must be screened for, as well as a safe swallow. Children with cerebral palsy often have sensory deficits or struggle with swallowing which makes them vulnerable to choking and aspiration, which could then lead to chest infections.

Activities

The following activities are examples of how elements of SI can be used to help treat sensory deficits depending on the sensory profile of the child.

It is important to consider:

- what state the child is currently in, so as to determine which activities are required to bring the child into a regulated state
- which activities would be **stimulating** to a child's system, thus increasing arousal, and which would be **inhibitory** and, thus, depressing to a child's system

> Note: Deep pressure touch, such as pushing hands into clay can be regulating, while light touch or stroking can increase arousal, and in some cases can even be interpreted as pain. Crinkly or rough textures increase arousal, while smooth and soft textures can be regulating.

> Note: Most proprioceptive activities are regulating, especially when bringing in rhythm and regularity.

> Note: Linear plane movements - back and forth and side-to-side can be calming, while quick, unpredictable movements or spinning can be excitatory. Up-and-down movements can also be excitatory, for example, bouncing on a trampoline or on a mother's lap.

The following strategies can be used for almost any sensory profile. They are used to regulate a child or, in other words, bring them into the 'just-right' state for learning and engagement:[2]

- **linear vestibular** - slow back and forth movement, e.g. on a swing or hammock.
- **deep pressure** - a hug offered by another (co-regulation), swaddling in a blanket etc.
- **low frequency vibration** - examples are a massage cushion or even driving in a car. For babies, it can be seen as holding them close and letting them feel the vibration of your chest as you talk to them. This vibration generally spans between a sound you can hear and something you can feel in your body. It is of low pitch, not high buzzing.
- **proprioception** - active movement of muscles. This is any movement against resistance or gravity. Often, a favourite for children is crawling through a spandex tunnel (offering a resistance against limb movement). Active play incorporating whole body movement is another example.

When considering the literature, it is apparent that there is a limited evidence base for treatment effectiveness. It has been highlighted that some studies have concluded that there are "...few positive outcomes and some null or negative outcomes". However, a limited evidence base does not mean an intervention is ineffective; rather subsequent validation may follow once large scale studies are conducted.[17]

- A 2013 systematic review,[30] updated in 2020,[31] concluded that sensory integration is a "red light therapy" for cerebral palsy, and is considered an ineffective treatment. This study further advised therapists to have open discussions with the family regarding "green light therapies", which have been shown to be effective and evidence-based.[30]
- However, further research studies have been conducted regarding the assessment of sensory deficits and SIT for children with cerebral palsy, especially as an adjunct to other conventional therapy. For example, a 2021 study showed that SIT in combination with conventional physiotherapy exercises were more effective than just conventional therapy and exercises in improving gross motor function in children with cerebral palsy.[32]

If parents and therapists decide to use SIT, they should establish clear, functional outcomes as the baseline before starting with therapy. This should be accompanied by parent, teacher and team member education. After 8-10 weeks, the outcome measures should be completed again to determine if the intervention is effective.[27]

Resources

- An overview of Sensory Integration theoretical models and application in the school setting: Understanding and Supporting Differences in Sensory Integration in the School Setting Part III
- What is Sensory Integration?
- Warutkar VB, Krishna Kovela R. Review of Sensory Integration Therapy for Children With Cerebral Palsy. Cureus. 2022 Oct 26;14(10):e30714.
- Case-Smith J, Weaver LL, Fristad MA. A systematic review of sensory processing interventions for children with autism spectrum disorders. Autism. 2015 Feb;19(2):133-48.
- Awalludin, ZA. Sensory Integration and Functional Movement: A Guide to Optimal Development in Early Childhood. 4th International Conference on Arts Language and Culture (ICALC 2019)
- Tracking Sensory Development

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